<u>ASSIGNMENT - 1</u>

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Que 1: Wireless Technologies and Characteristics

Everyday wireless technologies typically fall into local or wide-area access categories, each with distinct characteristics:

- <u>WiFi (IEEE 802.11):</u> This is a wireless local area network (WLAN) technology that provides a shared transmission rate of up to more than 100 Mb/ps. Users must typically be within a few 10s of meters of the access point (base station).
- <u>Cellular Networks (3G/4G LTE):</u> These provide wide-area wireless access, enabling users to be within a few tens of kilometers of a base station. LTE can achieve downstream rates in excess of 10 Mbps.
- <u>Bluetooth:</u> This technology operates over very short distances, such as one or two meters, and is often used for personal area networks linking devices like headsets or keyboards.

<u>Preference:</u> When given a choice, WiFi is generally preferred while stationary due to its higher throughput and shared local nature. Cellular networks are preferred when on the move because they provide wide-area mobility and do not require the user to stay within the small radius of a single access point.

Que 2: Link Sharing (2 Mbps Link)

a) In circuit switching, the resources needed along a path are reserved for the duration of the communication session. Since each user requires a constant 1 Mbps, a 2 Mbps link can support exactly 2 users simultaneously.

b) Packet Switching Queuing Delay

- <u>Two or fewer users:</u> In a packet-switched network, link use is allocated on demand. If two or fewer users transmit simultaneously, the maximum aggregate arrival rate is 2 Mbps (two users at 1 Mbps each). Because this rate does not exceed the link's 2 Mbps output capacity, packets will flow through the link essentially without delay.

- <u>Three users:</u> If three users transmit at the same time, the aggregate arrival rate becomes 3 Mbps. Because this exceeds the 2 Mbps output capacity of the link, the arriving packets must wait in the output buffer, resulting in a queuing delay.

c) Probability of a Single User Transmitting
Because each user transmits only 20% of the time, the probability (p) that a given user is transmitting at any specific moment is 0.2.

d) Three Users Transmitting Simultaneously
Assuming the users transmit independently, the probability that all three users transmit simultaneously is the product of their individual probabilities - 0.2 * 0.2 * 0.2 = 0.008
Therefore, the probability is 0.008 (or 0.8%). The queue only grows when the aggregate arrival rate exceeds the link capacity, which in this 3-user scenario only happens when all three users transmit at the same time. Thus, the fraction of time the queue grows is also 0.8%.

<u>Que 3: Packet Sizes in a Packet-Switched Internet</u>

<u>Advantage of Large Packets:</u> Sending larger packets reduces the proportion of overhead data. Every packet sent requires header bytes (such as IP and TCP headers). By placing more data into a single large packet, the overall ratio of header bytes to payload bytes decreases, leading to more efficient use of the link's transmission rate.

<u>Disadvantage of Large Packets:</u> Most packet switches use store-and-forward transmission, meaning the router must receive the entire packet before it can begin transmitting the first bit onto the outbound link. The transmission delay for a packet is L/R (where L is the packet length in bits and R is the transmission rate). Therefore, larger packets result in longer store-and-forward delays at every router along the path. Additionally, if a packet overflows a router's finite buffer or suffers bit-level corruption, the entire packet is lost and must be retransmitted. Retransmitting a massive packet is much more costly to network resources than retransmitting a small one.

Que 4: Exceeding Prescribed Cable Lengths

When physical media such as twisted-pair copper wire, coaxial cable, or optical fiber are extended beyond their prescribed lengths, several issues occur:

- Signal Attenuation: Environmental considerations and the physical medium itself cause path loss, which decreases the signal strength as it travels over a distance. Over extended lengths, the signal may degrade to the point where data cannot be accurately recovered.
- Increased Propagation Delay: The time required for a bit to travel from the beginning of the link to the end is the propagation delay, which is the distance divided by the propagation speed (d/s). Longer cables linearly increase this delay.

Que 5: Circuit Switching vs Packet Switching

The Difference:

- Circuit Switching: The resources (such as transmission rate and buffers) needed along a path are reserved for the duration of the communication session. This pre-allocates link use regardless of demand.
- Packet Switching: Resources are not reserved; a session's messages use resources on demand. Packets may have to wait in a queue for access to a communication link, allocating link use on a packet-by-packet basis.

Efficiency Scenarios:

- Packet Switching - more efficient for bursty traffic, where users alternate between periods of activity and inactivity. For example, when multiple users are Web surfing, they rarely request data at exactly the same time, allowing packet switching to support more users by sharing the transmission capacity on demand and avoiding wasted idle resources.
- Circuit Switching - more efficient for real-time services like telephone calls or video conferencing. These applications require a constant, guaranteed transmission rate and don’t suffer from the variable and unpredictable end-to-end queuing delays that occur in packet-switched networks.